

Variance Reduction Note 1: The Convergence of STORM

Alina Lodhi

Abstract

Stochastic recursive momentum (STORM) reduces gradient estimation error by combining a moving average with a correction evaluated at two consecutive iterates using the same random sample. This tutorial explains that mechanism and gives a self-contained convergence proof for a fixed-parameter, finite-horizon version with a single initialization batch. Under unbiasedness, bounded variance, and mean-square smoothness, the method attains an expected squared gradient norm of order $T^{-2/3}$ after T iterations, yielding $O(\varepsilon^{-3})$ stochastic gradient evaluations for an expected gradient norm at most ε , with problem parameters fixed. The analysis is pedagogical and makes no claim of a new algorithm or convergence rate. A selective reading list covers variance reduction, compositional optimization, sign-based optimization, and minimax problems.

1 Why recursive variance reduction?

Consider the smooth, possibly nonconvex problem

$$\min_{x \in \mathbb{R}^d} F(x) := \mathbb{E}_{\xi}[f(x; \xi)]. \quad (1)$$

Write $g(x; \xi) = \nabla f(x; \xi)$ for a stochastic gradient. Plain stochastic gradient descent uses $g(x_t; \xi_t)$ directly; even near a stationary point, its noise need not disappear. For general smooth nonconvex objectives, the standard bounded-variance analysis gives an $O(T^{-1/2})$ bound on the expected *squared* gradient norm with a suitable step size [1]. Variance reduction uses additional structure in the oracle to improve the dependence on the target accuracy.

In finite-sum optimization, gradient memories and control variates underlie SAG, SVRG, and SAGA [4, 6, 8]. Recursive estimators, including SARAH and SPIDER [23, 26], instead exploit the fact that the gradient changes gradually along an optimization trajectory. STORM, introduced by Cutkosky and Orabona [33], couples such gradient differences with a momentum average. Related hybrid estimators were developed by Tran-Dinh et al. [34].

This note isolates the resulting error recursion. We analyze a *fixed-parameter STORM recursion with batched initialization*; the original STORM algorithm uses an adaptive schedule and needs no batches [33]. Thus the theorem below is an explanatory specialization, not a reproduction of the original adaptive theorem. For a broader account of variance reduction, see Gower et al. [2].

2 The estimator and its assumptions

Assumption 1. *The function F is differentiable and bounded below by a finite number $F_{\star}$. There are constants $L > 0$ and $\sigma \geq 0$ such that, for every $x, y \in \mathbb{R}^d$,*

$$\mathbb{E}[g(x; \xi)] = \nabla F(x), \quad \mathbb{E} \|g(x; \xi) - \nabla F(x)\|^2 \leq \sigma^2, \quad (2)$$

$$\mathbb{E} \|g(x; \xi) - g(y; \xi)\|^2 \leq L^2 \|x - y\|^2. \quad (3)$$

Fresh samples are independent of the past, and the oracle permits evaluating the same sampled function at two specified points. All norms are Euclidean.

By Jensen's inequality, (3) implies that F is L -smooth. It is a condition on the *sampled gradients*, stronger than smoothness of F alone. The shared-sample oracle is essential to the following argument; two independent gradient calls cannot be substituted without changing the analysis.

Fix a deterministic x_0 , an integer horizon $T \geq 1$, a step size $\eta > 0$, a mixing parameter $a \in (0, 1]$, and an initialization size $B \geq 1$. Draw B independent samples and set

$$d_0 = \frac{1}{B} \sum_{i=1}^B g(x_0; \zeta_i). \quad (4)$$

For $t = 0, \dots, T-1$, perform

$$x_{t+1} = x_t - \eta d_t, \quad (5)$$

$$d_{t+1} = g(x_{t+1}; \xi_{t+1}) + (1-a)[d_t - g(x_t; \xi_{t+1})]. \quad (6)$$

The same fresh ξ_{t+1} is used in both terms of (6). Return x_R , where R is independent and uniform on $\{0, \dots, T-1\}$. This implementation uses $B + 2T$ stochastic gradient evaluations.

To interpret the update, rewrite it as

$$d_{t+1} = (1-a)d_t + ag(x_{t+1}; \xi_{t+1}) + (1-a)[g(x_{t+1}; \xi_{t+1}) - g(x_t; \xi_{t+1})]. \quad (7)$$

The last term corrects the old estimate for movement of the iterate. Its noise is controlled by $\|x_{t+1} - x_t\|$, while the coefficient a attenuates the noise introduced by the new gradient. The estimate d_t is generally conditionally biased: the proof tracks its mean-square error rather than assuming unbiasedness at each iteration.

3 A short convergence proof

Define the tracking error $e_t = d_t - \nabla F(x_t)$, and let $\mathcal{F}_t$ contain all samples used to construct x_t and d_t . In particular, x_{t+1} is $\mathcal{F}_t$ -measurable, and ξ_{t+1} is independent of $\mathcal{F}_t$. Write

$$q_t = \mathbb{E} \|e_t\|^2, \quad D_t = \mathbb{E} \|d_t\|^2, \quad G_t = \mathbb{E} \|\nabla F(x_t)\|^2.$$

Lemma 1 (Tracking error). *Under Assumption 1,*

$$q_{t+1} \leq (1-a)q_t + 2L^2\eta^2 D_t + 2a^2\sigma^2, \quad q_0 \leq \sigma^2/B. \quad (8)$$

Proof. Let $\delta(x; \xi) = g(x; \xi) - \nabla F(x)$. Subtracting $\nabla F(x_{t+1})$ from (6) gives

$$e_{t+1} = (1-a)e_t + (1-a)[\delta(x_{t+1}; \xi_{t+1}) - \delta(x_t; \xi_{t+1})] + a\delta(x_{t+1}; \xi_{t+1}).$$

Conditionally on $\mathcal{F}_t$, both noise terms have zero mean, so their cross terms with e_t vanish. The two noise terms need not be independent; we bound their squared sum by twice the sum of their squared norms. Moreover, the variance identity and (3) give

$$\begin{aligned} & \mathbb{E} [\|\delta(x_{t+1}; \xi_{t+1}) - \delta(x_t; \xi_{t+1})\|^2 \mid \mathcal{F}_t] \\ & \leq \mathbb{E} [\|g(x_{t+1}; \xi_{t+1}) - g(x_t; \xi_{t+1})\|^2 \mid \mathcal{F}_t] \leq L^2\eta^2 \|d_t\|^2. \end{aligned}$$

Consequently,

$$\mathbb{E} [\|e_{t+1}\|^2 \mid \mathcal{F}_t] \leq (1-a)^2 \|e_t\|^2 + 2(1-a)^2 L^2\eta^2 \|d_t\|^2 + 2a^2\sigma^2.$$

Take expectations and use $(1-a)^2 \leq 1-a$. The initialization bound follows from independence and zero mean of the B initial gradient errors. $\square$

Lemma 2 (Descent). *For the update (5),*

$$\mathbb{E}F(x_{t+1}) \leq \mathbb{E}F(x_t) - \frac{\eta}{2}G_t - \frac{\eta}{2}(1 - L\eta)D_t + \frac{\eta}{2}q_t. \quad (9)$$

Proof. Smoothness gives

$$F(x_{t+1}) \leq F(x_t) - \eta \langle \nabla F(x_t), d_t \rangle + \frac{L\eta^2}{2} \|d_t\|^2.$$

Substitute the identity

$$2 \langle \nabla F(x_t), d_t \rangle = \|\nabla F(x_t)\|^2 + \|d_t\|^2 - \|e_t\|^2$$

and take expectations. $\square$

Theorem 1 (A finite-horizon guarantee). *Suppose Assumption 1 holds, and write $\Delta = F(x_0) - F_\star$. If*

$$0 < \eta \leq \frac{1}{2L}, \quad a = 4L^2\eta^2,$$

then the recursion (4)–(6) satisfies

$$\mathbb{E} \|\nabla F(x_R)\|^2 \leq \frac{2\Delta}{\eta T} + \frac{\sigma^2}{aBT} + 2a\sigma^2. \quad (10)$$

In particular, choosing

$$\eta = \frac{1}{2LT^{1/3}}, \quad a = T^{-2/3}, \quad B = \lceil T^{1/3} \rceil \quad (11)$$

yields

$$\mathbb{E} \|\nabla F(x_R)\|^2 \leq \frac{4L\Delta + 3\sigma^2}{T^{2/3}}, \quad \mathbb{E} \|\nabla F(x_R)\| \leq \frac{\sqrt{4L\Delta + 3\sigma^2}}{T^{1/3}}. \quad (12)$$

Proof. Use the nonnegative Lyapunov quantity

$$\Phi_t = \mathbb{E}[F(x_t) - F_\star] + \frac{\eta}{2a}q_t.$$

Multiply (8) by $\eta/(2a)$ and add (9). The q_t terms cancel, leaving

$$\begin{aligned} \Phi_{t+1} - \Phi_t &\leq -\frac{\eta}{2}G_t + \left[-\frac{\eta}{2}(1 - L\eta) + \frac{L^2\eta^3}{a} \right] D_t + \eta a \sigma^2 \\ &= -\frac{\eta}{2}G_t + \left[-\frac{\eta}{4} + \frac{L\eta^2}{2} \right] D_t + \eta a \sigma^2 \leq -\frac{\eta}{2}G_t + \eta a \sigma^2. \end{aligned}$$

Here the equality uses $a = 4L^2\eta^2$, and the last inequality uses $L\eta \leq 1/2$. Sum over $t = 0, \dots, T-1$, use $\Phi_T \geq 0$, and note $\Phi_0 \leq \Delta + \eta\sigma^2/(2aB)$. Since $\mathbb{E} \|\nabla F(x_R)\|^2 = T^{-1} \sum_{t=0}^{T-1} G_t$, this proves (10). Substitution of (11), with $B \geq T^{1/3}$, gives the first bound in (12); Cauchy–Schwarz gives the second. $\square$

What the guarantee says. Taking $T \geq \max\{1, \lceil (4L\Delta + 3\sigma^2)^{3/2}\varepsilon^{-3} \rceil\}$ makes $\mathbb{E} \|\nabla F(x_R)\| \leq \varepsilon$. The oracle cost is $2T + \lceil T^{1/3} \rceil = O(\varepsilon^{-3})$ for fixed L, Δ, σ . The initialization cost is included. The argument assumes neither bounded gradient norms nor bounded function values above, but its schedule requires L and T . With $B = 1$, the same proof retains a $\sigma^2 T^{-1/3}$ initialization term in the squared-norm bound; one cannot simply remove the batch and keep this theorem unchanged. When $\sigma = 0$, instead choose $\eta = 1/(2L)$ and $a = 1$ in (10) to obtain $4L\Delta/T$.

The ε^{-3} exponent agrees with lower bounds in the appropriate mean-square-smooth stochastic oracle model [48]; this statement does not claim optimal dependence on every problem parameter. Adaptive STORM variants require their own schedules and analyses [35, 37, 39]. Likewise, the theorem does not directly cover compositions, sign updates, or saddle points: these introduce additional estimation errors or require different descent and stationarity measures.

4 Recommended reading

Variance-reduction methods. Basic guarantees [1–3]; gradient memories and control variates [4–10]; sampling and acceleration [11–19]; nonconvex and recursive estimators [20–32]; momentum and adaptive schedules [33–43]; second-order extensions [44–47]; lower bounds [48].

Compositional optimization. Two-level approximation [49–52]; finite-sum and recursive estimators [53–56]; multilevel tracking [57–60]; coupled and partially observed blocks [61–65]; projection-free methods [66–70]; statistical-model applications and federated extensions [71–74].

Sign-based optimization. Quantization, signSGD, majority voting, and error feedback [75–79]; variance-reduced sign updates and Lion [80–82]; compressed gradient tracking [83–87]. Related distributed control-variate methods include SCAFFOLD, FedPAGE, decentralized hybrid estimators, and ProxSkip [88–91].

Minimax optimization. Finite-sum saddle-point methods, matrix games, and variance-reduced extragradient [92–95]; recursive, hybrid, adaptive, and decentralized nonconvex minimax methods [96–99]; a variational-inequality treatment [100].

References

- [1] Saeed Ghadimi and Guanghui Lan. Stochastic first- and zeroth-order methods for nonconvex stochastic programming. *SIAM Journal on Optimization*, 23(4):2341–2368, 2013.
- [2] Robert M. Gower, Mark Schmidt, Francis Bach, and Peter Richtárik. Variance-reduced methods for machine learning. *Proceedings of the IEEE*, 108(11):1968–1983, 2020.
- [3] Eduard Gorbunov, Filip Hanzely, and Peter Richtárik. A Unified Theory of SGD: Variance Reduction, Sampling, Quantization and Coordinate Descent. In *AISTATS*, PMLR 108:680–690, 2020.
- [4] Nicolas Le Roux, Mark Schmidt, and Francis Bach. A stochastic gradient method with an exponential convergence rate for finite training sets. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 25, 2012.
- [5] Shai Shalev-Shwartz and Tong Zhang. Stochastic Dual Coordinate Ascent Methods for Regularized Loss Minimization. *Journal of Machine Learning Research*, 14(16):567–599, 2013.
- [6] Rie Johnson and Tong Zhang. Accelerating stochastic gradient descent using predictive variance reduction. In *NeurIPS*, volume 26, 2013.
- [7] Jakub Konecny and Peter Richtárik. Semi-Stochastic Gradient Descent Methods. *arXiv preprint arXiv:1312.1666*, 2013.
- [8] Aaron Defazio, Francis Bach, and Simon Lacoste-Julien. SAGA: A fast incremental gradient method with support for non-strongly convex composite objectives. In *NeurIPS*, volume 27, 2014.
- [9] Aaron J. Defazio, Tiberio S. Caetano, and Justin Domke. Finito: A Faster, Permutable Incremental Gradient Method for Big Data Problems. *arXiv preprint arXiv:1407.2710*, 2014.
- [10] Lin Xiao and Tong Zhang. A proximal stochastic gradient method with progressive variance reduction. *SIAM Journal on Optimization*, 24(4):2057–2075, 2014.
- [11] Hongzhou Lin, Julien Mairal, and Zaid Harchaoui. A Universal Catalyst for First-Order Optimization. In *NeurIPS*, volume 28, 2015.
- [12] Zeyuan Allen-Zhu. Katyusha: The first direct acceleration of stochastic gradient methods. *arXiv preprint arXiv:1603.05953*, 2016.

- [13] Zeyuan Allen-Zhu and Yang Yuan. Improved SVRG for Non-Strongly-Convex or Sum-of-Non-Convex Objectives. In *ICML*, PMLR 48:1080–1089, 2016.
- [14] Zeyuan Allen-Zhu. Katyusha X: Simple Momentum Method for Stochastic Sum-of-Nonconvex Optimization. In *ICML*, PMLR 80:179–185, 2018.
- [15] Xun Qian, Zheng Qu, and Peter Richtarik. SAGA with Arbitrary Sampling. In *ICML*, PMLR 97:5190–5199, 2019.
- [16] Robert M. Gower, Peter Richtarik, and Francis Bach. Stochastic Quasi-Gradient Methods: Variance Reduction via Jacobian Sketching. *arXiv preprint arXiv:1805.02632*, 2018.
- [17] Fanhua Shang, Yuanyuan Liu, Kaiwen Zhou, James Cheng, Kelvin Kai Wing Ng, and Yuichi Yoshida. Guaranteed Sufficient Decrease for Stochastic Variance Reduced Gradient Optimization. In *AISTATS*, PMLR 84:1027–1036, 2018.
- [18] Dmitry Kovalev, Samuel Horvath, and Peter Richtarik. Don’t Jump Through Hoops and Remove Those Loops: SVRG and Katyusha are Better Without the Outer Loop. *arXiv preprint arXiv:1901.08689*, 2019.
- [19] Xun Qian, Zheng Qu, and Peter Richtarik. L-SVRG and L-Katyusha with Arbitrary Sampling. *Journal of Machine Learning Research*, 22(112), 2021.
- [20] Sashank J. Reddi, Ahmed Hefny, Suvrit Sra, Barnabas Poczos, and Alex Smola. Stochastic variance reduction for nonconvex optimization. In *International Conference on Machine Learning (ICML)*, PMLR 48:314–323, 2016.
- [21] Zeyuan Allen-Zhu and Elad Hazan. Variance reduction for faster non-convex optimization. In *ICML*, PMLR 48:699–707, 2016.
- [22] Zeyuan Allen-Zhu. Natasha: Faster Non-Convex Stochastic Optimization via Strongly Non-Convex Parameter. In *ICML*, PMLR 70:89–97, 2017.
- [23] Lam M. Nguyen, Jie Liu, Katya Scheinberg, and Martin Takáč. SARAH: A novel method for machine learning problems using stochastic recursive gradient. In *ICML*, 2017.
- [24] Lam M. Nguyen, Katya Scheinberg, and Martin Takac. Inexact SARAH Algorithm for Stochastic Optimization. *arXiv preprint arXiv:1811.10105*, 2018.
- [25] Lihua Lei, Cheng Ju, Jianbo Chen, and Michael I. Jordan. Non-convex finite-sum optimization via SCSSG methods. In *NeurIPS*, volume 30, 2017.
- [26] Cong Fang, Chris Junchi Li, Zhouchen Lin, and Tong Zhang. SPIDER: Near-optimal non-convex optimization via stochastic path-integrated differential estimator. In *NeurIPS*, volume 31, 2018.
- [27] Dongruo Zhou, Pan Xu, and Quanquan Gu. Stochastic nested variance reduction for nonconvex optimization. In *NeurIPS*, volume 31, 2018.
- [28] Zhe Wang, Kaiyi Ji, Yi Zhou, Yingbin Liang, and Vahid Tarokh. SpiderBoost and momentum: Faster stochastic variance reduction algorithms. In *NeurIPS*, volume 32, 2019.
- [29] Nhan H. Pham, Lam M. Nguyen, Dzong T. Phan, and Quoc Tran-Dinh. ProxSARAH: An Efficient Algorithmic Framework for Stochastic Composite Nonconvex Optimization. *Journal of Machine Learning Research*, 21(110):1–48, 2020.
- [30] Zhize Li, Slavomir Hanzely, and Peter Richtarik. ZeroSARAH: Efficient Nonconvex Finite-Sum Optimization with Zero Full Gradient Computation. *arXiv preprint arXiv:2103.01447*, 2021.
- [31] Zhize Li, Hongyan Bao, Xiangliang Zhang, and Peter Richtárik. PAGE: A simple and optimal probabilistic gradient estimator for nonconvex optimization. In *ICML*, 2021.
- [32] Samuel Horvath and Peter Richtarik. Nonconvex Variance Reduced Optimization with Arbitrary Sampling. In *ICML*, PMLR 97:2781–2789, 2019.

- [33] Ashok Cutkosky and Francesco Orabona. Momentum-based variance reduction in non-convex SGD. In *NeurIPS*, volume 32, 2019.
- [34] Quoc Tran-Dinh, Nhan H. Pham, Dzung T. Phan, and Lam M. Nguyen. A hybrid stochastic optimization framework for stochastic composite nonconvex optimization. *arXiv preprint arXiv:1907.03793*, 2019.
- [35] Kfir Y. Levy, Ali Kavis, and Volkan Cevher. STORM+: Fully adaptive SGD with recursive momentum for nonconvex optimization. In *NeurIPS*, volume 34, 2021.
- [36] Feihu Huang, Junyi Li, and Heng Huang. SUPER-ADAM: Faster and Universal Framework of Adaptive Gradients. In *NeurIPS*, volume 34, 2021.
- [37] Zijian Liu, Ta Duy Nguyen, Thien Hang Nguyen, Alina Ene, and Huy L. Nguyen. META-STORM: Generalized fully-adaptive variance reduced SGD for unbounded functions. *arXiv preprint arXiv:2209.14853*, 2022.
- [38] Ali Kavis, Stratis Skoulakis, Kimon Antonakopoulos, Leello Tadesse Dadi, and Volkan Cevher. Adaptive Stochastic Variance Reduction for Non-convex Finite-Sum Minimization. In *NeurIPS*, volume 35, 2022.
- [39] Wei Jiang, Sifan Yang, Yibo Wang, and Lijun Zhang. Adaptive variance reduction for stochastic optimization under weaker assumptions. In *NeurIPS*, volume 37, pages 22047–22080, 2024.
- [40] Benjamin Dubois-Taine, Sharan Vaswani, Reza Babanezhad, Mark Schmidt, and Simon Lacoste-Julien. SVRG Meets AdaGrad: Painless Variance Reduction. *arXiv preprint arXiv:2102.09645*, 2021.
- [41] Bingcong Li, Lingda Wang, and Georgios B. Giannakis. Almost Tune-Free Variance Reduction. In *ICML*, PMLR 119:5969–5978, 2020.
- [42] Zijian Liu, Ta Duy Nguyen, Alina Ene, and Huy L. Nguyen. Adaptive Accelerated (Extra-)Gradient Methods with Variance Reduction. In *ICML*, PMLR 162:13947–13994, 2022.
- [43] Ganzhao Yuan. Adaptive Lipschitz-Free Proximal Gradient Methods for Stochastic Composite Nonconvex Optimization. *arXiv preprint arXiv:2502.21099v4*, 2026.
- [44] Zhize Li. SSRGD: Simple Stochastic Recursive Gradient Descent for Escaping Saddle Points. In *NeurIPS*, volume 32, 2019.
- [45] Zeyuan Allen-Zhu. Natasha 2: Faster Non-Convex Optimization Than SGD. In *NeurIPS*, volume 31, 2018.
- [46] Zhe Wang, Yi Zhou, Yingbin Liang, and Guanghui Lan. Stochastic Variance-Reduced Cubic Regularization for Nonconvex Optimization. In *AISTATS*, PMLR 89:2731–2740, 2019.
- [47] Dongruo Zhou, Pan Xu, and Quanquan Gu. Stochastic Variance-Reduced Cubic Regularized Newton Methods. In *ICML*, PMLR 80:5990–5999, 2018.
- [48] Yossi Arjevani, Yair Carmon, John C. Duchi, Dylan J. Foster, Nathan Srebro, and Blake Woodworth. Lower bounds for non-convex stochastic optimization. *Mathematical Programming*, 199:165–214, 2023.
- [49] Mengdi Wang, Ethan X. Fang, and Han Liu. Stochastic compositional gradient descent: Algorithms for minimizing compositions of expected-value functions. *arXiv preprint arXiv:1411.3803*, 2014.
- [50] Mengdi Wang, Ji Liu, and Ethan X. Fang. Accelerating stochastic composition optimization. *Journal of Machine Learning Research*, 18(105):1–23, 2017.
- [51] Saeed Ghadimi, Andrzej Ruszczyński, and Mengdi Wang. A Single Timescale Stochastic Approximation Method for Nested Stochastic Optimization. *SIAM Journal on Optimization*, 2020.
- [52] Tianyi Chen, Yuejiao Sun, and Wotao Yin. Solving Stochastic Compositional Optimization is Nearly as Easy as Solving Stochastic Optimization. *arXiv preprint arXiv:2008.10847*, 2020.

- [53] Xiangru Lian, Mengdi Wang, and Ji Liu. Finite-sum composition optimization via variance reduced gradient descent. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR 54:1159–1167, 2017.
- [54] Tsung-Yu Hsieh, Yasser El-Manzalawy, Yiwei Sun, and Vasant Honavar. Compositional Stochastic Average Gradient for Machine Learning and Related Applications. *arXiv preprint arXiv:1809.01225*, 2018.
- [55] Junyu Zhang and Lin Xiao. A composite randomized incremental gradient method. In *ICML*, PMLR 97:7454–7462, 2019.
- [56] Wenqing Hu, Chris Junchi Li, Xiangru Lian, Ji Liu, and Huizhuo Yuan. Efficient Smooth Non-Convex Stochastic Compositional Optimization via Stochastic Recursive Gradient Descent. In *NeurIPS*, volume 32, 2019.
- [57] Junyu Zhang and Lin Xiao. Multi-level composite stochastic optimization via nested variance reduction. *arXiv preprint arXiv:1908.11468*, 2019.
- [58] Wei Jiang, Bokun Wang, Yibo Wang, Lijun Zhang, and Tianbao Yang. Optimal algorithms for stochastic multi-level compositional optimization. In *ICML*, PMLR 162:10195–10216, 2022.
- [59] Krishnakumar Balasubramanian, Saeed Ghadimi, and Anthony Nguyen. Stochastic Multi-level Composition Optimization Algorithms with Level-Independent Convergence Rates. *arXiv preprint arXiv:2008.10526*, 2020.
- [60] Wei Jiang, Sifan Yang, Yibo Wang, Tianbao Yang, and Lijun Zhang. Revisiting stochastic multi-level compositional optimization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(7):5613–5624, 2025.
- [61] Bokun Wang and Tianbao Yang. Finite-Sum Coupled Compositional Stochastic Optimization: Theory and Applications. In *ICML*, PMLR 162:23292–23317, 2022.
- [62] Wei Jiang, Gang Li, Yibo Wang, Lijun Zhang, and Tianbao Yang. Multi-block-single-probe variance reduced estimator for coupled compositional optimization. In *NeurIPS*, volume 35, pages 32499–32511, 2022.
- [63] Quanqi Hu, Dixian Zhu, and Tianbao Yang. Non-Smooth Weakly-Convex Finite-sum Coupled Compositional Optimization. In *NeurIPS*, volume 36, 2023.
- [64] Wei Jiang, Sifan Yang, Yibo Wang, Lijun Zhang, and Zechao Li. Solving finite-sum coupled compositional optimization via multi-block-single-probe estimator. *arXiv preprint arXiv:2609.15723*, 2026.
- [65] Xingyu Chen, Bokun Wang, Ming Yang, Qihang Lin, and Tianbao Yang. Stochastic Momentum Methods for Non-smooth Non-Convex Finite-Sum Coupled Compositional Optimization. In *NeurIPS*, volume 38, 2025.
- [66] Elad Hazan and Haipeng Luo. Variance-Reduced and Projection-Free Stochastic Optimization. In *ICML*, PMLR 48:1263–1271, 2016.
- [67] Tesi Xiao, Krishnakumar Balasubramanian, and Saeed Ghadimi. A Projection-free Algorithm for Constrained Stochastic Multi-level Composition Optimization. In *NeurIPS*, volume 35, 2022.
- [68] Wei Jiang, Sifan Yang, Wenhao Yang, Yibo Wang, Yuanyu Wan, and Lijun Zhang. Projection-free variance reduction methods for stochastic constrained multi-level compositional optimization. In *ICML*, PMLR 235:21962–21987, 2024.
- [69] Hamed Hassani, Amin Karbasi, Aryan Mokhtari, and Zebang Shen. Stochastic Conditional Gradient++. *arXiv preprint arXiv:1902.06992*, 2019.
- [70] Wei Jiang, Sifan Yang, Wenhao Yang, Yibo Wang, Yuanyu Wan, Zechao Li, and Lijun Zhang. Projection-free multi-level algorithms for stochastic constrained compositional optimization. *arXiv preprint arXiv:2609.15679*, 2026.

- [71] Wei Jiang, Jiayu Qin, Lingyu Wu, Changyou Chen, Tianbao Yang, and Lijun Zhang. Learning unnormalized statistical models via compositional optimization. In *ICML*, PMLR 202:15105–15124, 2023.
- [72] Davoud Ataee Tarzanagh, Mingchen Li, Christos Thrampoulidis, and Samet Oymak. FedNest: Federated Bilevel, Minimax, and Compositional Optimization. In *ICML*, PMLR 162:21146–21179, 2022.
- [73] Wei Jiang, Jiayu Qin, Lingyu Wu, Changyou Chen, Tianbao Yang, and Lijun Zhang. Optimizing unnormalized statistical models through compositional optimization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 48(2):1949–1960, 2026.
- [74] Hongchang Gao, Junyi Li, and Heng Huang. On the Convergence of Local Stochastic Compositional Gradient Descent with Momentum. In *ICML*, PMLR 162:7017–7035, 2022.
- [75] Dan Alistarh, Demjan Grubic, Jerry Li, Ryota Tomioka, and Milan Vojnovic. QSGD: Communication-Efficient SGD via Gradient Quantization and Encoding. In *NeurIPS*, volume 30, 2017.
- [76] Wei Wen, Cong Xu, Feng Yan, Chunpeng Wu, Yandan Wang, Yiran Chen, and Hai Li. TernGrad: Ternary Gradients to Reduce Communication in Distributed Deep Learning. In *NeurIPS*, volume 30, 2017.
- [77] Jeremy Bernstein, Yu-Xiang Wang, Kamyar Azizzadenesheli, and Anima Anandkumar. signSGD: Compressed optimisation for non-convex problems. In *ICML*, PMLR 80:560–569, 2018.
- [78] Jeremy Bernstein, Jiawei Zhao, Kamyar Azizzadenesheli, and Anima Anandkumar. signSGD with majority vote is communication efficient and fault tolerant. In *International Conference on Learning Representations (ICLR)*, 2019.
- [79] Sai Praneeth Karimireddy, Quentin Rebjock, Sebastian U. Stich, and Martin Jaggi. Error feedback fixes SignSGD and other gradient compression schemes. In *ICML*, PMLR 97:3252–3261, 2019.
- [80] Wei Jiang, Sifan Yang, Wenhao Yang, and Lijun Zhang. Efficient sign-based optimization: Accelerating convergence via variance reduction. In *NeurIPS*, volume 37, pages 33891–33932, 2024.
- [81] Xiangning Chen, Chen Liang, Da Huang, Esteban Real, Kaiyuan Wang, Hieu Pham, Xuanyi Dong, Thang Luong, Cho-Jui Hsieh, Yifeng Lu, and Quoc V. Le. Symbolic discovery of optimization algorithms. In *NeurIPS*, volume 36, 2023.
- [82] Wei Jiang, Mao Xu, Wenhao Yang, Yibo Wang, Zechao Li, and Lijun Zhang. Convergence analysis of the Lion optimizer in centralized and distributed settings. In *ICML*, PMLR 306:53308–53333, 2026.
- [83] Konstantin Mishchenko, Eduard Gorbunov, Martin Takac, and Peter Richtarik. Distributed Learning with Compressed Gradient Differences. *arXiv preprint arXiv:1901.09269*, 2019.
- [84] Eduard Gorbunov, Konstantin P. Burlachenko, Zhize Li, and Peter Richtarik. MARINA: Faster Non-Convex Distributed Learning with Compression. In *ICML*, PMLR 139:3788–3798, 2021.
- [85] Wei Jiang, Zechao Li, and Lijun Zhang. Revisiting distributed sign-based variance reduction. *arXiv preprint arXiv:2609.18656*, 2026.
- [86] Peter Richtarik, Igor Sokolov, and Ilyas Fatkhullin. EF21: A New, Simpler, Theoretically Better, and Practically Faster Error Feedback. In *NeurIPS*, volume 34, 2021.
- [87] Alexander Tyurin and Peter Richtarik. DASHA: Distributed Nonconvex Optimization with Communication Compression, Optimal Oracle Complexity, and No Client Synchronization. *arXiv preprint arXiv:2202.01268*, 2022.
- [88] Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank Reddi, Sebastian Stich, and Ananda Theertha Suresh. SCAFFOLD: Stochastic Controlled Averaging for Federated Learning. In *ICML*, PMLR 119:5132–5143, 2020.

- [89] Haoyu Zhao, Zhize Li, and Peter Richtarik. FedPAGE: A Fast Local Stochastic Gradient Method for Communication-Efficient Federated Learning. *arXiv preprint arXiv:2108.04755*, 2021.
- [90] Ran Xin, Usman A. Khan, and Soumya Kar. A Hybrid Variance-Reduced Method for Decentralized Stochastic Non-Convex Optimization. In *ICML*, PMLR 139:11459–11469, 2021.
- [91] Konstantin Mishchenko, Grigory Malinovsky, Sebastian Stich, and Peter Richtarik. ProxSkip: Yes! Local Gradient Steps Provably Lead to Communication Acceleration! Finally!. In *ICML*, PMLR 162:15750–15769, 2022.
- [92] P. Balamurugan and Francis Bach. Stochastic variance reduction methods for saddle-point problems. In *NeurIPS*, volume 29, 2016.
- [93] Yair Carmon, Yujia Jin, Aaron Sidford, and Kevin Tian. Variance Reduction for Matrix Games. In *NeurIPS*, volume 32, 2019.
- [94] Tatjana Chavdarova, Gauthier Gidel, Francois Fleuret, and Simon Lacoste-Julien. Reducing Noise in GAN Training with Variance Reduced Extragradient. In *NeurIPS*, volume 32, 2019.
- [95] Junchi Yang, Siqi Zhang, Negar Kiyavash, and Niao He. A Catalyst Framework for Minimax Optimization. In *NeurIPS*, volume 33, 2020.
- [96] Luo Luo, Haishan Ye, Zhichao Huang, and Tong Zhang. Stochastic recursive gradient descent ascent for stochastic nonconvex-strongly-concave minimax problems. In *NeurIPS*, volume 33, 2020.
- [97] Quoc Tran-Dinh, Deyi Liu, and Lam M. Nguyen. Hybrid Variance-Reduced SGD Algorithms for Minimax Problems with Nonconvex-Linear Function. In *NeurIPS*, volume 33, 2020.
- [98] Feihu Huang, Xidong Wu, and Zhengmian Hu. AdaGDA: Faster adaptive gradient descent ascent methods for minimax optimization. In *AISTATS*, PMLR 206:2365–2389, 2023.
- [99] Lesi Chen, Haishan Ye, and Luo Luo. An Efficient Stochastic Algorithm for Decentralized Nonconvex-Strongly-Concave Minimax Optimization. In *AISTATS*, PMLR 238:1990–1998, 2024.
- [100] Ahmet Alacaoglu and Yura Malitsky. Stochastic variance reduction for variational inequality methods. In *Conference on Learning Theory (COLT)*, PMLR 178:778–816, 2022.